

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1007	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
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Date:	1/09/2026	Duration:	60 minutes

Code of Conduct

*I **Sk. Parvez Ahamed (192411084)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____Sk. Parvez Ahamed____

Blockchain-Based Academic Credential Verification Platform

1. Introduction

Academic certificates are important records used to prove a student's educational qualifications, but traditional verification methods often depend on centralized databases, manual checking, and direct communication with educational institutions. As the assigned problem statement, the Blockchain-Based Academic Credential Verification Platform aims to provide secure certificate issuance, tamper-resistant storage, fast verification, and reliable authentication of academic credentials. This report reviews existing research related to blockchain-based certificate verification, covering blockchain and smart contracts for credential management, IPFS-based decentralized storage, QR-code verification, and cryptographic techniques for detecting certificate tampering. Six papers published between 2020 and 2026 in credible journals and conferences were selected because each addresses a function required by the proposed platform: credential security, decentralized storage, certificate verification, and digital credential management. The papers are compared based on their blockchain approach, storage mechanism, verification method, security, scalability, and privacy, after which the report identifies the research gap most relevant to the assigned platform and discusses future directions.

2. Research Paper Selection and Review

Paper 1: Docschain: Blockchain-Based IoT Solution for Verification of Degree Documents

Rasool, Saleem, Iqbal, Dagiuklas, Mumtaz, and Qayyum (2020), published in *IEEE Transactions on Computational Social Systems*, proposed Docschain for secure and automated verification of degree documents. The objective is to reduce the time and cost involved in traditional degree verification while maintaining compatibility with existing document workflows. The system uses Optical Character Recognition (OCR) to extract information from degree documents, generates cryptographic hashes, and records verification information on a blockchain using Proof of Existence. The approach also supports bulk submission of previously issued degrees. This paper is directly relevant to the proposed platform because it demonstrates how blockchain can be

combined with document processing to authenticate academic certificates without requiring the complete document to be stored on the blockchain.

Paper 2: The Impact of the Blockchain on Academic Certificate Verification System—Review

Kumutha and Jayalakshmi (2021) reviewed the use of blockchain technology for academic certificate verification. The study focuses on the problems associated with paper-based certificates, including forgery, unauthorized modification, and slow manual verification. Blockchain, distributed ledgers, cryptographic hashing, and digital certificates are discussed as methods for improving certificate authenticity and security. The review concludes that blockchain can provide a tamper-resistant record and reduce dependence on centralized verification authorities. This paper is relevant to the proposed platform because it establishes the security and authenticity benefits of blockchain for academic credentials, although it provides limited experimental evaluation of a complete working system.

Paper 3: A Systematic Literature Review on Blockchain-Based Systems for Academic Certificate Verification

Rustemi, Dalipi, Atanasovski, and Risteski (2023), published in *IEEE Access*, conducted a systematic literature review of blockchain-based academic certificate verification systems. The study examined a large body of research and selected relevant studies to identify common technologies, architectures, benefits, and challenges. The review found that blockchain can improve certificate security, transparency, authenticity, and verification efficiency. However, it also identified unresolved issues involving scalability, privacy, interoperability, standardization, and institutional adoption. This paper is particularly relevant to the proposed platform because it provides a broad overview of existing academic credential systems and clearly identifies the major research gaps that need to be addressed.

Paper 4: Blockchain-Based Micro-Credentialing System in Higher Education Institutions: Systematic Literature Review

Alsobhi, Alakhtar, Ubaid, Hussain, and Hussain (2023), published in *Knowledge-Based Systems*, reviewed blockchain applications for micro-credentialing in higher education. The study

investigates how blockchain can support portable, verifiable, and trustworthy educational credentials beyond traditional degree certificates. The reviewed approaches use blockchain and digital credential technologies to represent skills, courses, and other educational achievements. The findings indicate that blockchain can improve the portability and authenticity of credentials while reducing the difficulty of verification. However, real-world adoption and interoperability remain limited. This paper is relevant to the proposed platform because it shows how the system could be extended beyond degree certificates to support training certificates, skill credentials, and micro-credentials.

Paper 5: Educational Certificate Verification System: Enhancing Security and Authenticity using Ethereum Blockchain and IPFS

Jaafar, Alsaad, and Al-Kabi (2024) developed a decentralized educational certificate verification system using Ethereum blockchain and IPFS. The system stores certificate-related information on the blockchain while using IPFS for decentralized document storage. Smart contracts are used to manage certificate records, and QR codes allow users to quickly access verification information. The approach improves certificate authenticity and reduces dependence on centralized databases. This paper is directly relevant to the proposed platform because the combination of Ethereum, IPFS, smart contracts, and QR codes provides a practical architecture for secure certificate issuance and verification.

Paper 6: Blockchain-Based Academic Certificate Verification System

Deshmukh, Thulkar, Somwanshi, Nimbalkar, and Jagadale (2026) proposed a blockchain-based academic certificate verification system using Ethereum, IPFS, React, Node.js, Ethers.js, Solidity, and QR codes. The system separates certificate storage from blockchain records by storing documents through IPFS and maintaining verification information on the blockchain. The study evaluates certificate verification and system performance under multiple users. The results demonstrate the feasibility of using blockchain for academic credential authentication while also highlighting issues related to transaction cost, verification latency, scalability, privacy, and dependence on external storage. This paper is highly relevant to the proposed platform because it represents a recent practical implementation combining decentralized storage, smart contracts, and a web-based verification interface.

3. Comparative Analysis

Table 1 compares the six reviewed papers across the aspects most relevant to the Blockchain-Based Academic Credential Verification Platform: blockchain approach, storage and verification technologies, security, scalability, privacy, main contribution, and limitation.

Paper (Author, Year)	Blockchain / Verification Approach	Storage / Tools	Security / Scalability / Privacy	Main Contribution	Limitation
Rasool et al. (2020)	Blockchain + Proof of Existence	OCR, hashing, blockchain	High / Moderate / Limited	Automated degree verification	OCR dependency
Kumutha & Jayalakshmi (2021)	Blockchain-based certificate authentication	Digital certificates, hashing	High / Not evaluated / Limited	Identifies blockchain benefits for certificate security	Mainly review-based
Rustemi et al. (2023)	Systematic review of blockchain credential systems	Blockchain, smart contracts	High / Major challenge / Major challenge	Identifies research trends and gaps	Based on existing literature

Paper (Author, Year)	Blockchain / Verification Approach	Storage / Tools	Security / Scalability / Privacy	Main Contribution	Limitation
Alsobhi et al. (2023)	Blockchain micro- credentialing	Digital credentials, blockchain	High / Limited / Moderate	Supports portable educational credentials	Limited real- world adoption
Jaafar et al. (2024)	Ethereum smart- contract verification	Ethereum, IPFS, QR codes	High / Moderate / Limited	Practical decentralized certificate verification	Cost and scalability concerns
Deshmukh et al. (2026)	Ethereum + IPFS certificate verification	React, Node.js, Solidity, IPFS, QR	High / Tested / Limited	Recent end-to- end academic credential system	Cost, latency, privacy issues

The comparison shows a clear progression in the literature. The earlier research primarily establishes the potential of blockchain for certificate security and authentication, while later studies introduce practical architectures using smart contracts, IPFS, and QR codes. The systematic review by Rustemi et al. identifies scalability, privacy, interoperability, and standardization as major unresolved challenges. Recent implementation studies demonstrate that blockchain-based verification is technically feasible, but they still operate with limitations related to transaction cost, latency, privacy, and institutional integration. No single reviewed system completely addresses all these issues simultaneously.

4. Critical Analysis and Research Gap

Across the reviewed papers, blockchain-based systems demonstrate that academic credentials can be made more secure and easier to verify. Rasool et al. (2020) demonstrate automated document verification using OCR, hashing, and Proof of Existence, while Jaafar et al. (2024) and Deshmukh et al. (2026) demonstrate practical systems using Ethereum, IPFS, smart contracts, and QR codes. However, these systems generally focus on certificate authenticity and verification rather than providing a complete, scalable, privacy-aware credential management platform.

Beyond the verification gap, several issues remain. Many systems store or expose credential-related information without sufficiently addressing privacy. Public blockchain transaction costs and confirmation times can also become problematic when thousands of certificates are issued. Furthermore, most systems do not provide comprehensive credential revocation mechanisms or interoperability with different universities and existing academic information systems. These limitations restrict the deployment of such systems on a large institutional scale.

Research Gap

The main research gap is the absence of a single scalable and privacy-aware platform that combines academic credential issuance, decentralized storage, tamper detection, QR-based verification, credential revocation, and interoperability while minimizing blockchain storage and transaction costs. Existing studies address individual components such as blockchain authentication, IPFS storage, QR verification, or micro-credentials, but they do not completely integrate these capabilities into one practical platform. The proposed Blockchain-Based Academic Credential Verification Platform addresses this gap by combining secure credential generation, cryptographic hashing, decentralized document storage, smart-contract-based verification, and user-friendly QR-based authentication within a unified system.

5. Future Directions and Relevance

Building on the gaps identified above, several directions are relevant to the assigned problem. Using Layer-2 blockchain networks or permissioned blockchains could reduce transaction costs

and improve scalability. Storing only certificate hashes and essential metadata on-chain while keeping documents in encrypted IPFS or other decentralized storage can improve efficiency and privacy. A credential revocation mechanism should allow institutions to mark certificates as revoked without modifying the original blockchain record. Integration with W3C Verifiable Credentials and Decentralized Identifiers (DIDs) could improve interoperability between universities and employers. Privacy-preserving methods such as selective disclosure and zero-knowledge proofs could also allow users to prove their qualifications without unnecessarily exposing personal information. Finally, mobile and QR-based verification can make the platform more accessible to employers and other authorized verifiers.

6. Conclusion

The reviewed literature shows that blockchain, smart contracts, cryptographic hashing, IPFS, and QR codes can significantly improve the security and authenticity of academic credentials. Existing systems successfully demonstrate decentralized certificate issuance and verification, but challenges remain in scalability, privacy, transaction cost, revocation, interoperability, and large-scale institutional deployment. This gap directly supports the proposed Blockchain-Based Academic Credential Verification Platform, which aims to combine secure credential management, decentralized storage, fast verification, and privacy-aware architecture in a single system. The comparative analysis and identified research gap provide a literature-based foundation for the design and development of the proposed platform.

7. References

- Rasool, S., Saleem, A., Iqbal, M., Dagiuklas, T., Mumtaz, S., & Qayyum, Z. U. (2020). Docschain: Blockchain-based IoT solution for verification of degree documents. *IEEE Transactions on Computational Social Systems*, 7(3), 827–837.
- Kumutha, K., & Jayalakshmi, S. (2021). The impact of the blockchain on academic certificate verification system—Review. *EAI Endorsed Transactions on Energy Web*.

- Rustemi, A., Dalipi, F., Atanasovski, V., & Risteski, A. (2023). A systematic literature review on blockchain-based systems for academic certificate verification. *IEEE Access*, 11, 64679–64696.
- Alsobhi, H. A., Alakhtar, R. A., Ubaid, A., Hussain, O. K., & Hussain, F. K. (2023). Blockchain-based micro-credentialing system in higher education institutions: Systematic literature review. *Knowledge-Based Systems*, 265, 110238.
- Jaafar, R. A., Alsaad, S. N., & Al-Kabi, M. N. (2024). Educational certificate verification system: Enhancing security and authenticity using Ethereum blockchain and IPFS. *Al-Mustansiriyah Journal of Science*, 35(1).
- Deshmukh, R., Thulkar, P., Somwanshi, P., Nimbalkar, S., & Jagadale, N. (2026). Blockchain-based academic certificate verification system. *International Journal for Research in Applied Science and Engineering Technology*.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15 %)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15 %)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25